QRAFT

Quantum-Resistant
Anonymous Frame Transactions

Use one unified private balance to send funds anonymously.

Built with ERC-8286 PQ wallets and EIP-8141 transactions.

Tornado Cash Classic: limitations

Withdrawals depend on a centralised relayer	In the relayer route, a third-party service submits your transaction.
Paying another wallet requires a withdrawal	Funds leave the private pool and arrive at a public address.
Direct RPC requests expose your IP	Onchain privacy does not hide your network identity.
Funds are split into fixed-value notes	Each deposit has its own note; exact payments are awkward.
No built-in post-quantum wallet	Classic does not provide PQ account signatures or receiving encryption.

What QRAFT adds

One unified private balance	Automatic note selection, exact amounts and private change.
No application relayer	EIP-8141 goes directly through Anon RPC to the alternative mempool. Paymaster sponsors setup and gas.
Pay privately without withdrawing	Encrypted payments move between users inside the pool.
IP protection through Nym	Anon RPC routes reads and submissions through the mixnet.
ERC-8286 PQ wallets	ML-DSA account signatures and ML-KEM receiving encryption.

QRAFT and Tornado Cash Classic

Feature	Tornado Cash Classic	QRAFT
Payment flow	Fixed-denomination deposit and withdrawal	Exact private transfers and public withdrawals
Amounts and change	Withdraw a denomination, less applicable fees	Variable amounts; encrypted change notes
Note management	Deposit notes; optional encrypted Note Account	Automatic discovery, input selection and balance
Receiving funds	Withdrawal to a public recipient address	Encrypted payment to a QRAFT receiving address
Execution	Pool deposit and proof-based withdrawal calls	Sponsored EIP-8141 frame transactions

Scope: Classic only. Tornado Cash Nova already supports arbitrary amounts, shielded transfers and a shielded balance. This is a feature comparison, not a security ranking.

Sources: Classic deposit / withdrawal documentation · Nova documentation

Demo: a private ETH payment

Follow Alice and Bob through one payment.

01 · ALICE

Deposit 0.00004 ETH

Two deposits of 0.00002 ETH appear as one private balance.

02 · ALICE → BOB

Send 0.00003 ETH

Use Bob’s receiving address. Alice retains 0.00001 ETH privately.

03 · BOB

Receive 0.00003 ETH

Bob’s device discovers the encrypted payment and updates his balance.

04 · BOB

Withdraw 0.00001 ETH

Send to a public address. Bob retains 0.00002 ETH privately.

Switch to the app. Show confirmed results; label shortened waiting periods “Time compressed”.

How a private transfer works

01

Sender’s device

Selects notes, encrypts payment and change, generates a local proof.

→

02

Anon RPC + node

Browser submits EIP-8141 frames through Nym. Onchain contracts authorize and sponsor execution.

→

03

Private pool

Verifies proof and nullifiers, records commitments and encrypted notes.

→

04

Recipient’s device

Decrypts its notes, excludes spent notes and calculates the balance.

Sponsored setup and deposit

Factory deploys the ERC-8286 PQ account; paymaster funds stake and gas. Deposit ETH into the pool.

Withdraw from the pool

Proof → pool sends ETH to public address.

Deposit and withdrawal amounts and addresses are public. Private-transfer amounts are hidden by the proof and encrypted notes.

Implemented and tested

Verified flow

93 unit tests pass. Full browser flow through Nym passed: deposit, private send, recovery and withdrawal.

Cryptography

PQ account signatures and receiving encryption. Current Groth16 proofs are conventional cryptography.

Network privacy

Browser reads and payment submissions use Nym Anon RPC. No application relayer. Timing and onchain correlation remain possible.

Prototype status

Development proving setup. No independent security audit or production-security claim.

github.com/Shoydon/QRAFT · Implementation and test evidence